

Cross-Camera Evidence Correlation for Flagged Vehicles

Using Spatio-Temporal Reasoning over a Camera Topology Graph

1 Purpose and Scope

The existing single-camera pipeline detects a range of traffic violations independently at each camera site: overspeeding, helmet and seatbelt non-compliance, triple riding, stop-line and red-light violations, illegal parking, and wrong-side driving. Each of these detections is generated, stored, and reported as an isolated event tied to one camera at one point in time. This document describes the design of a complementary system that sits above the existing per-camera pipeline. Its purpose is to determine, once a vehicle has been flagged by any single camera, whether the same vehicle is later observed by other cameras in the network, so that all evidence relating to that vehicle across the entire camera network can be consolidated into a single case file rather than remaining scattered across independent camera logs.

The central idea is that a flagged vehicle's future position in the camera network can be predicted from the known physical layout of the road network and historical traffic behaviour, without requiring an expensive or unreliable city-wide visual search. Once a likely time window and a likely set of downstream cameras have been identified, a lightweight confirmation step (number-plate matching, with appearance matching as a fallback) is used to verify that the vehicle observed at a downstream camera is in fact the same vehicle that was originally flagged.

2 Camera Topology Graph

2.1 Structure

The camera network is represented as a graph whose nodes are the physical camera installations and whose edges represent direct, traversable road connections between two cameras. An edge is drawn between two cameras only if a vehicle can travel from one to the other without passing through a third monitored camera in between. Each edge carries a weight that represents the expected travel time, in seconds, for a vehicle travelling along that specific road segment between the two cameras.

In the reference implementation, three cameras, denoted Cam1, Cam2, and Cam3, are connected as follows: an edge of weight 40 connects Cam1 and Cam2, an edge of weight 60 connects Cam2 and Cam3, and an edge of weight 50 connects Cam1 and Cam3 directly. The presence of a direct edge between Cam1 and Cam3, in addition to the path through Cam2, reflects a real situation in which more than one physical road allows a vehicle to travel between two points, and each such road is represented as its own edge rather than being collapsed into a single shortest-path value. This is important because a vehicle leaving Cam1 may legitimately reach Cam3 by either route, and the system must be able to search for it at the times implied by both possibilities rather than only the faster one.

The edge weights are not fixed physical constants such as distance divided by speed limit. They

are derived from a model trained on historical transit-time distributions for each camera pair, incorporating the road distance between the two cameras as well as the number of intersections, signals, or other points of interruption that lie on that specific stretch of road. Two camera pairs that are the same straight-line distance apart can have different edge weights if one route passes through more signal-controlled junctions than the other, since each such junction introduces additional, variable delay. The edge weight stored in the graph should therefore be understood as a mean transit time, μ , with an associated spread, σ , learned from past traffic rather than from a single theoretical calculation.

2.2 Role of the Number-Plate Detector Camera

In addition to the cameras that participate in the general topology graph, the network includes a designated number-plate detector camera, referred to here as CamX to avoid confusion with the graph node already named Cam1 in the reference diagram. CamX is the camera at which the system attempts to read the number plate of a candidate vehicle once a time search window for that vehicle has been computed. CamX may correspond physically to any node in the topology graph; it is distinguished in this document purely by its functional role as the point at which plate confirmation is attempted, not by a separate physical location.

3 Shared State

All cameras in the network write to and read from a common shared state store. The shared state holds, for each actively tracked vehicle, a record analogous to the one already produced by the existing per-camera pipeline: a vehicle identifier, the set of violation flags raised so far, the camera and timestamp at which each flag was raised, and, once available, the vehicle's number plate. The shared state is the mechanism by which a flag raised at one camera becomes visible to the logic that computes time search windows for other cameras, and by which a confirmed plate match at a downstream camera is written back so that the case file continues to accumulate evidence rather than being treated as a new, unrelated event.

4 End-to-End Pipeline

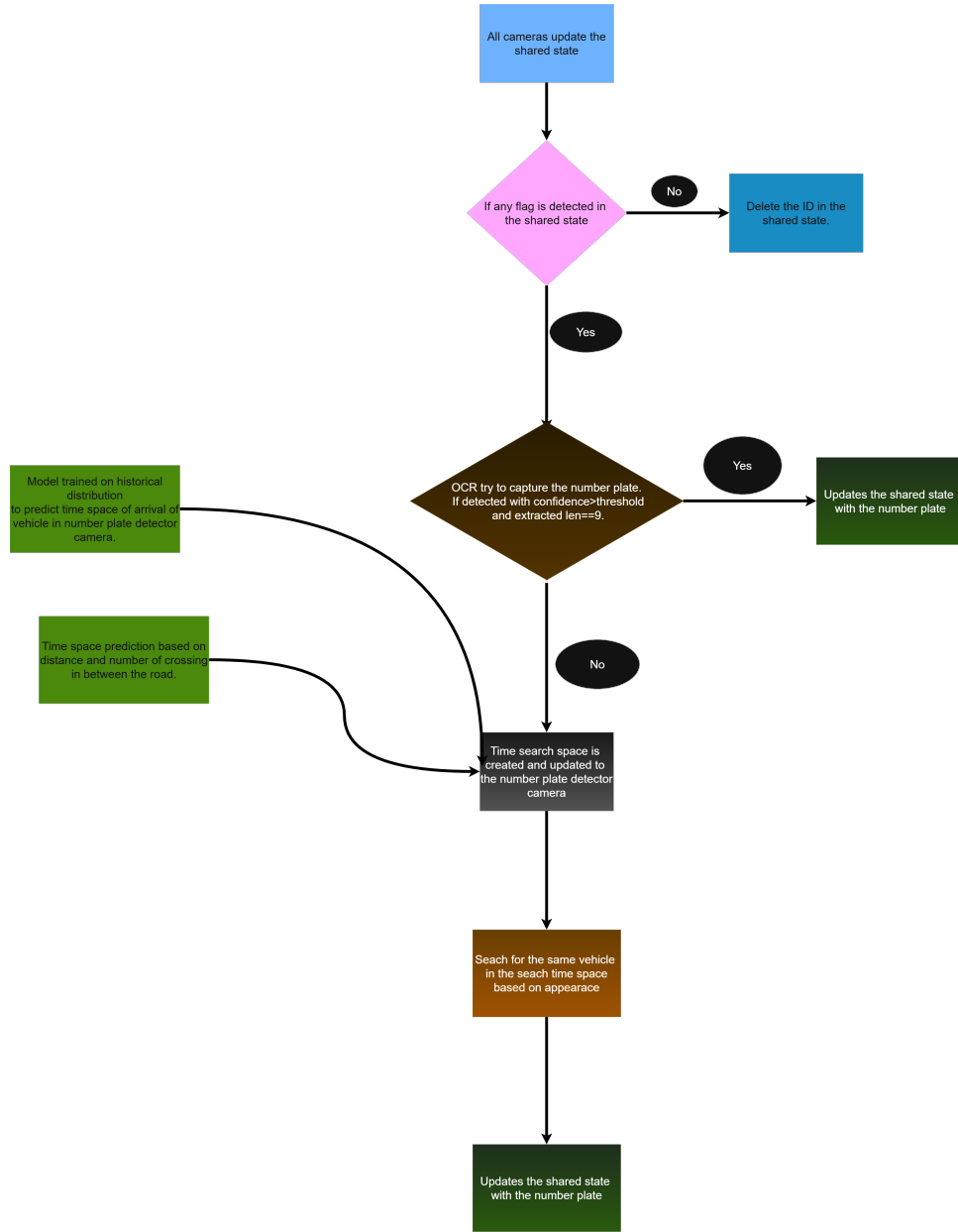


Figure 1: Workflow of evidence extractor.

The pipeline begins at the point where the existing per-camera violation detectors have already run. From there, the following sequence is followed.

4.1 Step 1: Shared State Update

Every camera, after completing its own local violation detection, writes its findings into the shared state. This is a continuous process and is not contingent on a violation having occurred; a vehicle with no violations is also written so that, if it is flagged later by a different camera further down its route, its earlier appearance is already available.

4.2 Step 2: Flag Check

The shared state is checked for the presence of any violation flag attached to a given vehicle identifier. If no flag is present, the vehicle’s entry is deleted from the shared state. This deletion step prevents the shared state from growing indefinitely with vehicles that present no evidentiary interest, since the purpose of the downstream pipeline is specifically to build a cross-camera case file for vehicles that have already been flagged for at least one violation. If a flag is present, the vehicle proceeds to the next step.

4.3 Step 3: Time Search Space Construction

For a flagged vehicle, the system constructs a time search space describing when and where the vehicle is expected to next appear in the camera network. This is done in two parts.

First, the distance-based component of the prediction is computed using the camera topology graph described in Section 2: the road distance between the camera that raised the flag and each reachable downstream camera, together with the number of intersections lying on that road segment, is used to estimate a baseline transit time for each candidate edge.

Second, this baseline is passed through a model trained on the historical transit-time distribution for that specific camera pair, which converts the baseline estimate into an expected arrival window rather than a single arrival instant. The result is, for each reachable downstream camera, a time interval during which the flagged vehicle is expected to arrive, centred on the historically observed mean transit time and widened according to the historically observed variance for that route. This interval is what is referred to as the time search space, and it is this interval, rather than a single predicted timestamp, that is written and updated against the number-plate detector camera, CamX, that lies on the relevant route.

4.4 Step 4: Number-Plate Confirmation at the Search Window

When CamX’s own clock falls within the time search space written for a given vehicle, the system attempts to read the number plate of vehicles passing CamX during that window using the existing OCR component of the pipeline. A read is accepted only if the OCR confidence exceeds a defined threshold and the extracted string has the expected length of nine characters, corresponding to a standard number-plate format. If a plate is successfully read and accepted, it is written into the shared state against the vehicle identifier, linking the new sighting at CamX to the original flagged record.

If no plate is successfully read and accepted within the time search space, the system does not immediately discard the candidate. Instead, it falls back to an appearance-based search: the candidates observed at CamX within the time search space are compared against the appearance of the originally flagged vehicle, and the same shared-state update is performed if an appearance match is found. This fallback exists specifically to handle the cases in which plate OCR fails due to poor viewing angle, occlusion, low light, or a damaged or obscured plate, which are common in practice and would otherwise cause a genuine match to be lost.

4.5 Step 5: Consolidation

Once the shared state for a vehicle identifier contains a confirmed plate (or appearance) match at a downstream camera, that camera’s evidence is merged into the same case file as the originally flagged violation. Over time, as a flagged vehicle continues to move through the network, this process repeats at each subsequent camera reachable from its current position, so that the final case file accumulates every relevant piece of evidence gathered across the network, rather than each camera’s findings remaining isolated.

5 Worked Example of the Time Search Space

The following example illustrates why the time search space is expressed as a window rather than a single predicted time, using the edge between two cameras two kilometres apart with a posted speed limit of forty kilometres per hour.

Historical logging of ordinary, non-flagged traffic along this specific route, collected over several weeks by the existing tracking and OCR pipeline, yields a mean transit time of four minutes with a standard deviation of one minute. This already establishes that even unflagged, typical traffic does not take a fixed time to cover this distance; some vehicles are delayed by a signal along the way, while others pass through more quickly.

A naive point-estimate approach would compute distance divided by speed limit, giving three minutes, and would check the downstream camera’s footage only at the single instant three minutes after the flag was raised. This approach fails in practice for two reasons: a vehicle delayed by a signal may arrive several minutes later than this estimate, and a vehicle that was itself flagged for speeding may arrive earlier than the speed-limit-based estimate would suggest, since it is, by definition, not travelling at the legal limit.

The window-based approach instead uses the historical mean and standard deviation directly. Taking a window of mean plus or minus two standard deviations around a flag raised at fourteen hundred thirty hours produces a search interval from fourteen-thirty-two to fourteen-thirty-six. Searching the downstream camera’s records across this four-minute interval, rather than at a single instant, captures the vehicle whether it travelled faster, slower, or was delayed by a signal, while still keeping the search space small enough to be practical. The choice of how many standard deviations to include controls a direct trade-off: a wider window increases the chance of catching the correct vehicle but also increases the number of irrelevant vehicles that must be filtered out by the plate or appearance confirmation step, while a narrower window is cleaner but risks missing the vehicle entirely on days when traffic conditions deviated from the historical norm.

6 Why Confirmation Is Required in Addition to Timing

The time search space narrows the set of vehicles that must be checked at a downstream camera, but it does not, by itself, identify which vehicle within that window is the one originally flagged, since other unrelated vehicles will ordinarily also pass through the same camera during the same interval. For this reason, the pipeline always pairs the time search space with a confirmation step. Number-plate matching is used as the primary confirmation signal because the existing pipeline already performs plate OCR as part of its standard evidence-gathering process, and a plate match is a strong, low-ambiguity identifier. Appearance-based matching is retained as a fallback specifically for the cases in which OCR confidence or extracted plate length fails to meet the required threshold, ensuring that a genuine match is not lost solely because of an imperfect plate read.

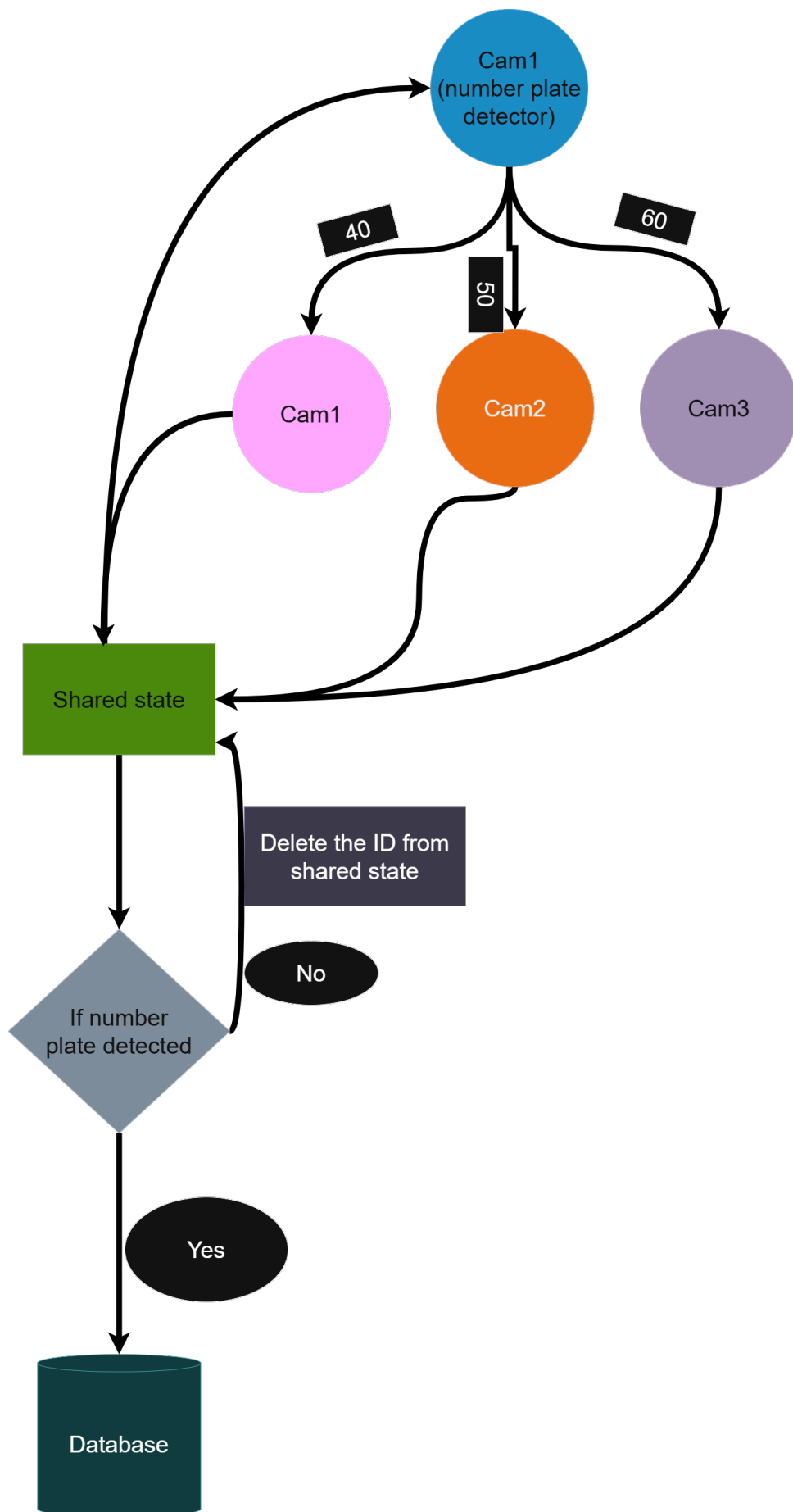


Figure 2: graph of evidence extractor.

7 Summary

The system extends the existing per-camera violation pipeline with a network-level layer that links a flagged vehicle’s evidence across multiple cameras. This is achieved by representing the physical camera network as a topology graph with empirically derived, time-distributed edge weights; using that graph, together with a flag raised in the shared state, to compute a time search space at each reachable downstream camera rather than a single predicted arrival time; and confirming candidate matches within that search space using number-plate OCR as the primary signal and appearance similarity as a fallback. The result is a single, consolidated case file per flagged vehicle that draws together every violation and every camera sighting relevant to that vehicle across the network, rather than a set of disconnected, single-camera records.